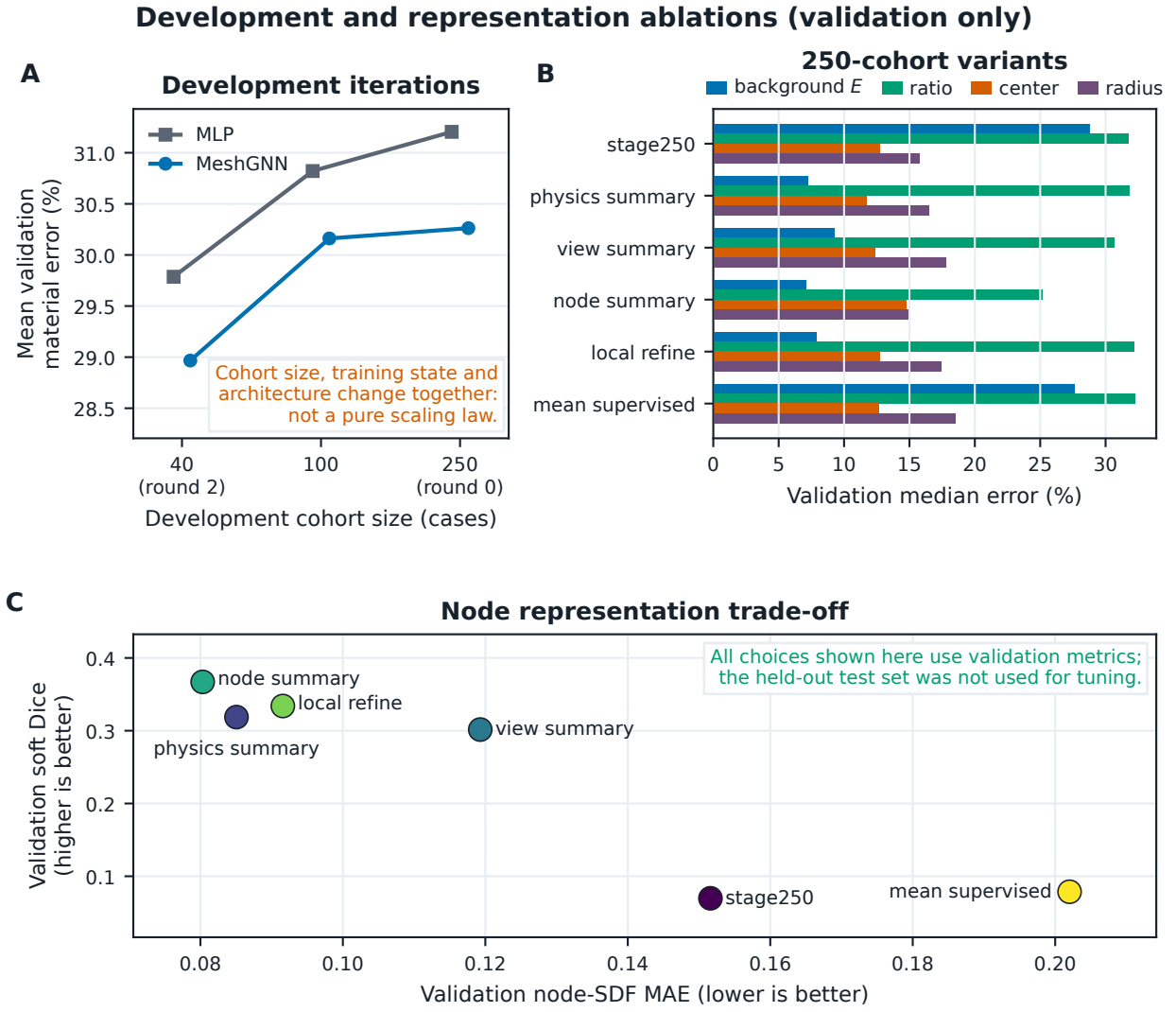
Development and representation ablations (validation only)
A
Development iterations
MLP
MeshGNN
Mean validation material error (%)
31.0
30.5
30.0
29.5
29.0
28.5
Cohort size, training state and architecture change together: not a pure scaling law.
40
(round 2)
100
250
(round 0)
Development cohort size (cases)
B
250-cohort variants
background E
ratio
center
radius
stage250
physics summary
view summary
node summary
local refine
mean supervised
0
5
10
15
20
25
30
Validation median error (%)
C
Node representation trade-off
All choices shown here use validation metrics; the held-out test set was not used for tuning.
Validation soft Dice (higher is better)
0.4
0.3
0.2
0.1
node summary
local refine
physics summary
view summary
stage250
mean supervised
0.08
0.10
0.12
0.14
0.16
0.18
0.20
Validation node-SDF MAE (lower is better)